

Bonus Booklet

Reinforcement Learning — CentraleSupélec

CentraleSupélec

This booklet collects exercises pulled out of the weekly TD sheets to keep those sheets short — typically the more abstract or notation-heavy problems, still worth doing but not essential to the session’s core exercise. One chapter per lecture; a chapter grows as exercises are added over the following sessions. **Exercise numbers below are inherited from the exercise’s original TD sheet**, not renumbered here, so it stays easy to see where a problem came from — they do not run consecutively within a chapter.

Chapter 1

Lecture 1 — MDP, Return, Value Functions, Bellman Equations

Pulled from TD 1’s main sheet (`../01-introduction-mdp/ exercises.tex`), whose own numbering has since been compacted to Exercises 1–11. References below to “the main sheet” point there.

Exercise 9 — Deriving the Bellman equations (theory bonus, Sutton & Barto Ch. 3)

The main sheet’s Exercise 6 *used* the Bellman expectation equation; this one **derives** it, from the definition of the return and nothing else. Throughout, π is an arbitrary policy and all expectations are taken over trajectories generated by π .

- (a) **From v_π down to q_π .** Starting from $v_\pi(s) \doteq \mathbb{E}_\pi[G_t | S_t = s]$ and conditioning on the action taken at time t , prove

$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a | s) q_\pi(s, a).$$

Hint: law of total expectation, splitting on the value of A_t ; the probability of $A_t = a$ given $S_t = s$ is precisely $\pi(a | s)$.

- (b) **From q_π down to v_π .** Using the recursion $G_t = R_{t+1} + \gamma G_{t+1}$ and conditioning this time on the pair (S_{t+1}, R_{t+1}) , prove

$$q_\pi(s, a) = \sum_{s', r} p(s', r | s, a) [r + \gamma v_\pi(s')].$$

At which exact step do you need the **Markov property**? State precisely what it is used for.

- (c) **Assembling.** Substitute (b) into (a) to recover the Bellman expectation equation for v_π — the one the main sheet’s Exercise 6 took for granted. Then substitute (a), written at s' , into (b) to obtain the Bellman expectation equation for q_π , i.e. $q_\pi(s, a)$ in terms of the $q_\pi(s', a')$ of its successors.
- (d) **Optimality.** Admit that an optimal policy exists. Show that $v_*(s) = \max_{a \in \mathcal{A}(s)} q_*(s, a)$, and explain why the $\sum_a \pi(a | s)$ of (a) becomes a $\max_a$ here, while the sum over (s', r) in (b) does **not**. Deduce the Bellman optimality equation for v_* .

Exercise 12 — The v_* / q_* / π_* dictionary (Sutton & Barto 3.25–3.28)

Four one-line equations, used constantly from Lecture 2 on. Give each, and justify it in a sentence.

- (a) v_* in terms of q_* .
- (b) q_* in terms of v_* and the four-argument $p(s', r \mid s, a)$.
- (c) π_* in terms of q_* .
- (d) π_* in terms of v_* and the four-argument p .
- (e) **The point.** Compare (c) and (d): to act optimally from q_* , what must you know about the environment? And from v_* ? Deduce why the algorithms in the second half of this course estimate q rather than v .

Exercise 13 — The same equations in three-argument notation (Sutton & Barto 3.29)

Much of the literature uses not the four-argument $p(s', r \mid s, a)$, but the pair

$$p(s' \mid s, a) \doteq \sum_{r \in \mathcal{R}} p(s', r \mid s, a), \quad r(s, a) \doteq \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a] = \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}^+} p(s', r \mid s, a).$$

- (a) Show that for any function v ,

$$\sum_{s', r} p(s', r \mid s, a) [r + \gamma v(s')] = r(s, a) + \gamma \sum_{s'} p(s' \mid s, a) v(s').$$

Which property of the bracket makes this work?

- (b) Deduce the four Bellman equations — for v_π , q_π , v_* and q_* — written with $p(s' \mid s, a)$ and $r(s, a)$ only.
- (c) In one sentence: what is lost, and what is gained, in this notation?

Exercise 14 — Episodic or continuing? (Sutton & Barto 3.5 and 3.6)

- (a) The normalization condition of the main sheet's Exercise 1(c),

$$\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) = 1 \quad \text{for all } s \in \mathcal{S}, a \in \mathcal{A}(s),$$

is written for the **continuing** case. Give the (very slight) modification needed for an **episodic** task, and say why it is needed. *Hint:* where does the agent land on the last step of an episode?

- (b) Pole-balancing, treated as an **episodic** task but *with* discounting: all rewards are 0 except -1 upon failure. Let K be the number of steps from t until the next failure. Give G_t .
- (c) The same task in the **continuing** formulation (the pole is reset after each failure and the task never ends). Give G_t now, and say how it differs from (b).
- (d) Check in both cases that maximizing the return really does mean “balance the pole as long as possible”.